DSA06 - Data Handling and Visualization
Activity 9: Map Storytelling Challenge

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Course Code / Course Name	DSA0602 / Data Handling and Visualization
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Scenario: You are building a map story about High-school graduates (% of adults, 1970) across the United States for a school-board meeting. The comparison variable is Illiteracy (% of population, 1970). The focus region is the West. The map uses 6 classes and the "cividis" colour scale. Statistics use all 50 states; the map shows the 48 contiguous states (Alaska and Hawaii are not in map_data). Story Brief MS-D06 (Tool: R, ggplot2 + maps).

Set-up: the code below was run first in R without any change.

```
library(ggplot2); library(maps)
d <- data.frame(state = state.name, zone = state.region, state.x77)
names(d) <- c("state", "zone", "pop", "income", "illit", "life", "murder", "hsgrad", "frost",
"area")
d$id <- tolower(d$state)
mapdat <- merge(map_data("state"), d, by.x = "region", by.y = "id")
mapdat <- mapdat[order(mapdat$group, mapdat$order), ]
V <- d$hsgrad      # the variable you map
W <- d$illit       # the comparison variable
```

Good map stories: one clear message, honest classes, rates not raw counts, a readable legend, and a source note.

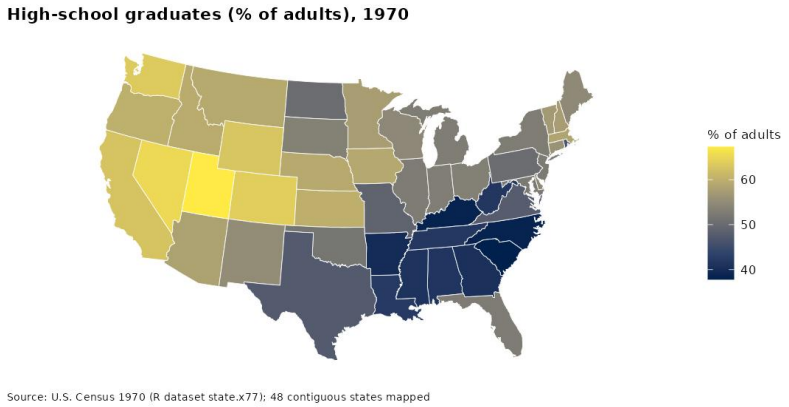
TASK 1: The base map

Draw a choropleth of hsgrad: ggplot(mapdat, aes(long, lat, group = group, fill = hsgrad)) + geom_polygon(colour = "white") + coord_map("albers", lat0 = 29.5, lat1 = 45.5) + scale_fill_viridis_c(option = "cividis") with a clear title and legend. Report the highest state and its value, the lowest value and the state(s) that have it, and the median of all 50 states.

```
ggplot(mapdat, aes(long, lat, group = group, fill = hsgrad)) +
  geom_polygon(colour = "white") +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_c(option = "cividis") +
  labs(title = "High-school graduates (% of adults), 1970",
       fill = "% of adults",
       caption = "Source: U.S. Census 1970 (R dataset state.x77)") +
  theme_void()

d[which.max(V), c("state", "hsgrad")]              # highest state
d[which(V == min(V)), c("state", "hsgrad")]        # lowest state
median(V)                                         # median of all 50 states
```

OUTPUT:



Output 1: High-school graduates (% of adults), 1970 (continuous cividis scale)

Measure (all 50 states)	Value
Highest state	Utah, 67.3%
Lowest value and state	37.8%, South Carolina (no other state has this value)
Median of all 50 states	53.25%

Ans:

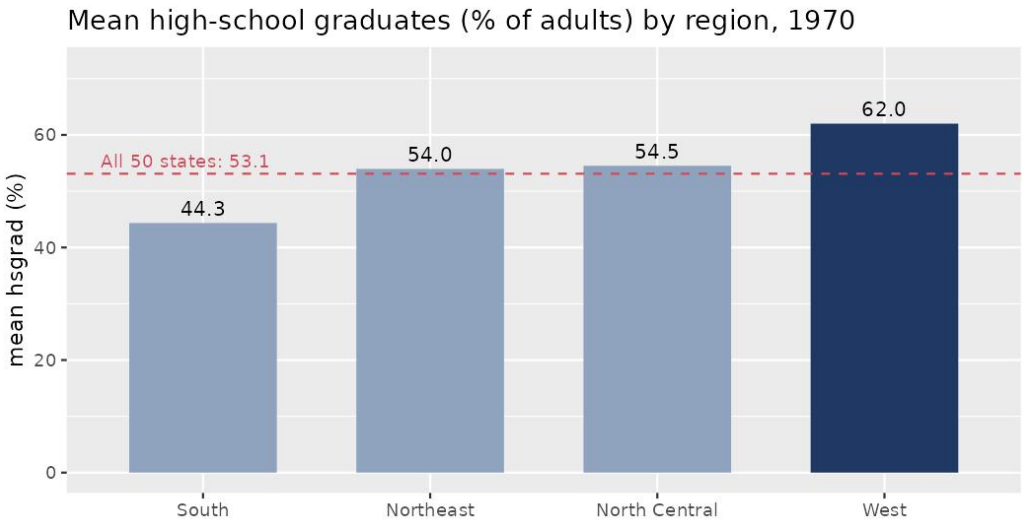
Utah has the highest share of adult high-school graduates (67.3%) and South Carolina the lowest (37.8%); the median of the 50 states is 53.25%. The map shows a clear pattern: the West is bright yellow, while a dark band runs through the South from Kentucky and North Carolina to Louisiana.

TASK 2: The regional story

Use tapply(V, d$zone, mean). Give the mean for each of the four regions and name the highest and lowest region and the gap between them. Then compute cor(V, W), describe its strength (weak below 0.3, moderate 0.3 to 0.7, strong above 0.7) and direction in one sentence, noting that it is not proof of cause.

```
tapply(V, d$zone, mean)        # mean for each region
cor(V, W)                       # hsgrad vs illiteracy
```

OUTPUT:



Output 2: Mean high-school graduates (% of adults) by region

Region	Mean hsgrad (%)	Note

Region	Mean hsgrad (%)	Note
Northeast	54.0	9 states
South	44.3	16 states, the lowest region
North Central	54.5	12 states
West	62.0	13 states, the highest region

Measure	Value
Gap between highest (West) and lowest (South)	62.0 − 44.3 = 17.7 percentage points
cor(V, W): hsgrad vs illiteracy	−0.66

Ans:

The West has the highest regional mean (62.0%) and the South the lowest (44.3%), a gap of 17.7 percentage points. The correlation with illiteracy is r = −0.66, which is moderate and negative: states with more high-school graduates tend to have lower illiteracy, but this is not proof that one causes the other.

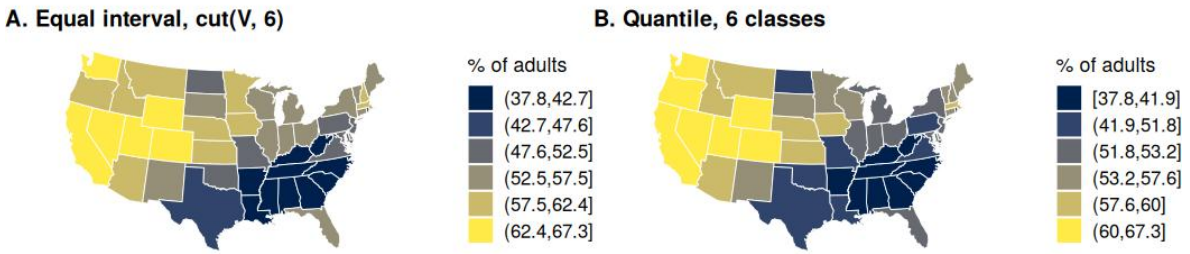
TASK 3: Classes change the story

Make two versions of the map with 6 classes: equal interval, cut(V, 6), and quantile, cut(V, quantile(V, seq(0, 1, length.out = 7)), include.lowest = TRUE). For each, report the lower limit of the top class and the number of states in it. Which method makes the highest values look rarer, and which would you choose for your audience?

```
d$eq <- cut(V, 6)                                          # equal interval
d$qb <- cut(V, quantile(V, seq(0, 1, length.out = 7)),
            include.lowest = TRUE)                        # quantile
levels(d$eq); table(d$eq)
levels(d$qb); table(d$qb)
quantile(V, seq(0, 1, length.out = 7))

mapdat$eq <- d$eq[match(mapdat$region, d$id)]
mapdat$qb <- d$qb[match(mapdat$region, d$id)]
ggplot(mapdat, aes(long, lat, group = group, fill = eq)) +    # repeat with fill = qb
  geom_polygon(colour = "white") +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_d(option = "cividis") + theme_void()
```

OUTPUT:



Output 3: The same data in 6 classes: (A) equal interval and (B) quantile

Method	Top class	Lower limit	States in top class

Method	Top class	Lower limit	States in top class
Equal interval	(62.4, 67.3]	62.4	7
Quantile	(60.0, 67.3]	60.0 (exactly 59.98)	9

Method	Number of states in classes 1 to 6 (lowest to highest)
Equal interval	10, 2, 7, 13, 11, 7
Quantile	9, 8, 8, 9, 7, 9

Ans:

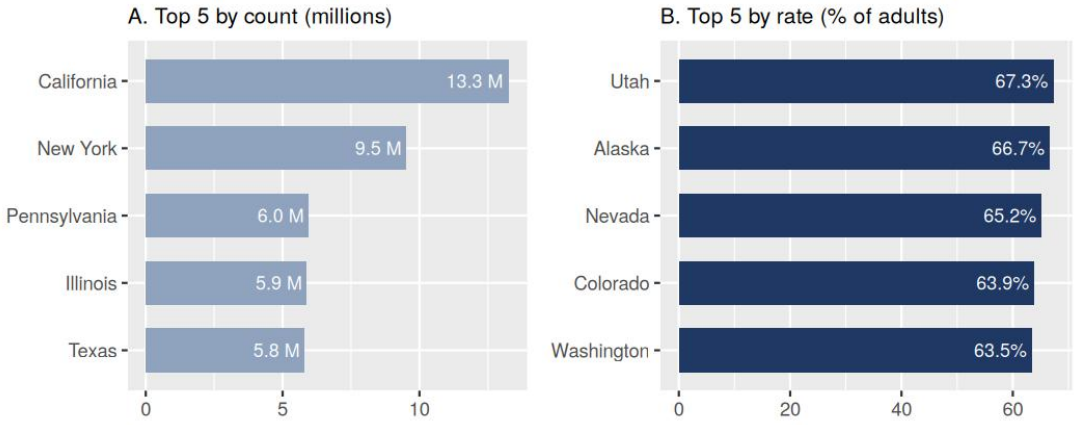
Equal interval makes the highest values look rarer: its top class starts at 62.4 and holds only 7 states, and its second class holds just 2 states, whereas the quantile top class starts at 60.0 and holds 9 states. For a school-board audience I would choose the quantile map, because each colour covers a similar number of states (7 to 9), so the map shows the ranking of states without an almost empty class; I would print the class limits in the legend so nobody mistakes the classes for equal-size steps.

TASK 4: Counts versus rates

Create the number of high-school graduates: count <- d$pop * 1000 * d$hsgrad / 100. Name the top 3 states by count and the top 3 states by hsgrad, say how many states appear in both lists, and in one sentence explain why a map of counts would mislead and what to map instead.

```
count <- d$pop * 1000 * d$hsgrad / 100
d$count <- count
head(d[order(-d$count), c("state", "count")], 3)     # top 3 by count
head(d[order(-d$hsgrad), c("state", "hsgrad")], 3)   # top 3 by rate
intersect(head(d$state[order(-d$count)], 3),
          head(d$state[order(-d$hsgrad)], 3))        # states in both lists
```

OUTPUT:



Output 4: Top 5 states by count of graduates (A) and by rate (B)

Rank	Top 3 by count	Count of graduates	Top 3 by hsgrad (%)
1	California	13,269,948	Utah, 67.3
2	New York	9,526,052	Alaska, 66.7
3	Pennsylvania	5,953,720	Nevada, 65.2

Ans:

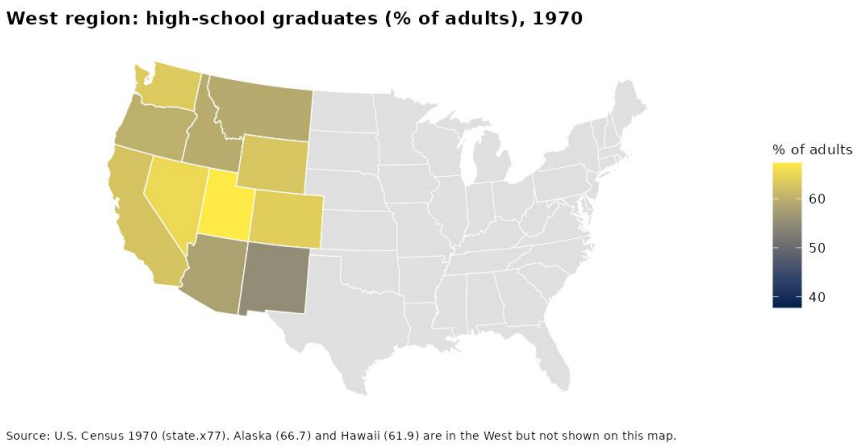
The top 3 by count are California, New York and Pennsylvania, and the top 3 by hsgrad are Utah, Alaska and Nevada, so 0 states appear in both lists. A map of counts would mislead because it would mostly show where the most people live (California, New York) and not where adults are best educated, so we should map the rate (% of adults) instead.

TASK 5: The story for the West region

Report the mean of hsgrad in West and its ratio to the mean of all 50 states, the highest state in West with its value, and how many West states are above the national median. Then write a headline of at most 12 words, and two evidence sentences using your numbers plus one recommendation for a school-board meeting.

```
west <- d$zone == "West"
mean(V[west])                          # West mean
mean(V)                                # national mean
mean(V[west]) / mean(V)                # ratio
d[west, ][which.max(d$hsgrad[west]), c("state", "hsgrad")]   # highest in West
sum(V[west] > median(V))               # West states above the national median
```

OUTPUT:



Output 5: The West region in focus (the other states in grey)

Measure	Value
Mean of hsgrad in the West	62.0%
Mean of all 50 states	53.1%
Ratio West ÷ all 50 states	62.0 ÷ 53.1 = 1.17
Highest state in the West	Utah, 67.3%
West states above the national median (53.25)	13 of 13

Part of the story	Text
Headline (9 words)	West leads the nation: high-school graduation 17% above average
Evidence 1	The 13 West states average 62.0% of adults with a high-school diploma, 1.17 times the national mean of 53.1%, and all 13 are above the national median of 53.25%.
Evidence 2	Utah is the highest at 67.3%, and the South, at 44.3%, trails the West by 17.7 points; states with more graduates also have less illiteracy (r = −0.66).
Recommendation	The board should study what Utah (67.3%) and Alaska (66.7%) do to keep students in school, and give extra support to the South, where fewer than half of adults finished high school.

Source note for the slide: U.S. Census 1970 (R dataset state.x77); rates are % of adults; the map shows the 48 contiguous states, while Alaska and Hawaii are included in the statistics.

Summary

Task	Topic	Finding

Task	Topic	Finding
1	Base map	Highest: Utah 67.3%; lowest: South Carolina 37.8%; median of 50 states 53.25%
2	Regional story	West 62.0% (highest), South 44.3% (lowest), gap 17.7 points; r(hsgrad, illiteracy) = −0.66 (moderate, negative)
3	Classes	Equal interval: top class from 62.4, 7 states. Quantile: top class from 60.0, 9 states. Chosen: quantile
4	Counts vs rates	Top by count: California, New York, Pennsylvania. Top by rate: Utah, Alaska, Nevada. 0 states in both
5	West story	West mean 62.0% = 1.17 × national mean; highest Utah 67.3%; 13 of 13 above the median

Good map story rule	How this map story follows it
One clear message	The West has the highest high-school graduation rates in 1970
Honest classes	Quantile classes, with class limits printed in the legend
Rates, not raw counts	% of adults is mapped, not the number of graduates
Readable legend	Titled legend (% of adults) with the colour-blind-friendly cividis scale
Source note	U.S. Census 1970 (state.x77) stated on the map

Conclusion

A good map story needs more than a coloured map. The same data looked different with equal interval and quantile classes (7 against 9 states in the top class), so the class method must be chosen and stated. Counts and rates gave completely different leaders (California against Utah, with no state in both top-3 lists), so the rate should be mapped. With these choices the message is clear and honest: in 1970 the West had the highest high-school graduation rate (62.0%, 1.17 times the national mean), and the South the lowest (44.3%).